Forma polar del oscilador bidimensional

Friday, August 28, 2026 2:53 AM

$$\mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{1}{2} \kappa (x^2 + y^2)$$

$x = r\cos\varphi \qquad y = r\sin\varphi \qquad , r(t), \varphi(t)$

$\dot{x} = \dot{r}\cos\varphi - r\sin\varphi \cdot \dot{\varphi}$ ← a)

$\dot{y} = \dot{r}\sin\varphi + r\cos\varphi \cdot \dot{\varphi}$

$$\dot{x}^2 + \dot{y}^2 = (\dot{r}\cos\varphi - r\sin\varphi \cdot \dot{\varphi})^2 + (\dot{r}\sin\varphi + r\cos\varphi \cdot \dot{\varphi})^2$$

$$= (\dot{r}\cos\varphi)^2 - 2\dot{r}\cos\varphi \cdot r\sin\varphi \cdot \dot{\varphi} + (r\sin\varphi \cdot \dot{\varphi})^2 + (\dot{r}\sin\varphi)^2 + 2\dot{r}\sin\varphi \cdot r\cos\varphi \cdot \dot{\varphi} + (r\cos\varphi \cdot \dot{\varphi})^2$$

▲ se anulan

$$= \dot{r}^2\cos^2\varphi + r^2\sin^2\varphi \cdot \dot{\varphi}^2 + \dot{r}^2\sin^2\varphi + r^2\cos^2\varphi \cdot \dot{\varphi}^2$$

$$= \dot{r}^2(\cos^2\varphi + \sin^2\varphi) + r^2\dot{\varphi}^2(\sin^2\varphi + \cos^2\varphi)$$

$$= \dot{r}^2 + r^2\dot{\varphi}^2$$

$$x^2 + y^2 = r^2\cos^2\varphi + r^2\sin^2\varphi = r^2$$

$$\mathcal{L} = \frac{1}{2} m (\dot{r}^2 + r^2\dot{\varphi}^2) - \frac{1}{2} \kappa r^2$$

b)

$\frac{\partial \mathcal{L}}{\partial \varphi} = 0 \qquad \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = cst = p_\varphi \qquad \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = mr^2 \cdot \dot{\varphi}$, la coordenada cíclica es φ

$x = r\cos\varphi \qquad \dot{x} = \dot{r}\cos\varphi - r\sin\varphi \cdot \dot{\varphi}$

$y = r\sin\varphi \qquad \dot{y} = \dot{r}\sin\varphi + r\cos\varphi \cdot \dot{\varphi}$

$$x\dot{y} = r\cos\varphi(\dot{r}\sin\varphi + r\cos\varphi \cdot \dot{\varphi})$$

$$= r\cos\varphi\dot{r}\sin\varphi + r^2\cos^2\varphi \cdot \dot{\varphi}$$

$$y\dot{x} = r\sin\varphi(\dot{r}\cos\varphi - r\sin\varphi \cdot \dot{\varphi})$$

$$= r\sin\varphi \cdot \dot{r}\cos\varphi - r^2\sin^2\varphi \cdot \dot{\varphi}$$

$$x\dot{y} - y\dot{x} = r\cos\varphi \cdot \dot{r}\sin\varphi + r^2\cos^2\varphi \cdot \dot{\varphi} - r\sin\varphi \cdot \dot{r}\cos\varphi + r^2\sin^2\varphi \cdot \dot{\varphi}$$

$$= r^2\cos^2\varphi \cdot \dot{\varphi} + r^2\sin^2\varphi \cdot \dot{\varphi}$$

$$= r^2\dot{\varphi} \qquad m(x\dot{y} - y\dot{x}) = mr^2\dot{\varphi}$$

c) $\frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r} \qquad p_\varphi = mr^2\dot{\varphi} \to kg\,m^2 \cdot \frac{rad}{s^2}$

$\frac{\partial \mathcal{L}}{\partial \dot{r}} = kg \cdot \left(\frac{m}{s}\right)$

d) $p_\varphi = mr^2\dot{\varphi} \qquad \mathcal{L} = \frac{1}{2} m (\dot{r}^2 + r^2\dot{\varphi}^2) - \frac{1}{2} \kappa r^2$

$\frac{\partial \mathcal{L}}{\partial r} = mr\dot{\varphi}^2 - \kappa r \qquad \frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r} \qquad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) = m\ddot{r}$

$\frac{\partial \mathcal{L}}{\partial \varphi} = 0 \qquad \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = mr^2\dot{\varphi} \qquad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\varphi}}\right) = 0$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial q_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0$$

$m\ddot{r} - mr\dot{\varphi}^2 + \kappa r = 0 \qquad p_\varphi = mr^2\dot{\varphi}$

$p_\varphi^2 = m^2r^4\dot{\varphi}^2$

$\frac{p_\varphi^2}{mr^3} = mr^3\dot{\varphi}^2$

$m\ddot{r} = mr\dot{\varphi}^2 - \kappa r$

$m\ddot{r} = \frac{p_\varphi^2}{mr^3} - \kappa r \qquad r = r_0$

$\dot{r} = 0$

$m\ddot{r} = \frac{p_\varphi^2}{mr^3} - \kappa r_0 \qquad \ddot{r} = 0$

$0 = \frac{p_\varphi^2}{mr_0^3} - \kappa r_0$

$\kappa r_0 = \frac{p_\varphi^2}{mr_0^3}$

$r_0^4 = \frac{p_\varphi^2}{m\kappa}$

$r_0 = \left(\frac{p_\varphi^2}{m\kappa}\right)^{1/4}$

Si $p_\varphi = 0$

$m\ddot{r} = -\kappa r \to$ Oscilador armónico simple.